COMP 451
Introduction to Natural Language Processing
Fall 2025

Project#2

Due Date: December 22, 2025 – 23:59 pm

I) Description:

In this project, you are expected to explore the use of prompt engineering for data annotation, focusing on four NLP tasks: i) Named Entity Recognition (NER), ii) Aspect-Based Sentiment Analysis (ABSA), iii) Coreference Resolution (COREF), and iv) Question Answering (Q/A). For the task that you select, follow the steps given below:

i) Find an annotated English or Turkish dataset from the web *(Dataset1)*

ii) Take 100-150 samples from the dataset (Dataset1), obtain its raw version and annotate it by using different prompt style configurations (i.e., zero-shot, few-shot, and chain-of-thought prompting). Use either large-scale LLMs (e.g., ChatGPT and Gemini), small-scale LLMs (e.g., Gemma, Llama, and Mistral), or Turkish-specific LLMs (e.g., Cosmos). Prompts should be written in English for multilingual models but should be written in Turkish for Turkish-specific models.

iii) Select four language models for your evaluations.

iv) Compare the annotations that you obtain using each LLM & prompt style configuration with the original annotations and report your similarity/success rate.

v) Collect raw data from the web (web scraping) and annotate them (50-100 samples) by using the best LLM & prompt style configuration with the highest success rate. *(Dataset2).*

vi) Format Dataset 2 in JSON format.

- <u>The classes for NER:</u> Person, Location, Organization, Time, Currency
- <u>The number of aspects for ABSA:</u> At least 4 aspects
- <u>The question types for Q/A:</u> Who, What, Where, When
- For evaluating the success rates, you can use publicly available NLP libraries (e.g., NLTK and spaCy)
- Research and analyze existing studies in the literature that are related to your topic
- Form groups of 3-4 students

II) Submissions:

Submit a single zip file (one submission per grup) to the Blackboard System (before due date) that includes:

- **Web Scraping Code:** Provide the code used for crawling data
- **Raw and Annotated Dataset 2:** Provide your dataset in JSON format

- **Final Report:** Write a report (at most 5 pages long) that shows the final prompts that are used for annotation, the web sites used for data collection, the description of how the crawler can be run, the success/similarity rate of prompts on Dataset 1.

IV) Late Policy:
Up to two days; -30 points the first day and -50 points the second day.

V) Plagiarism:
If plagiarism is detected, you will not receive any grade from the project.

VI) Grading:
<u>Dataset: & Web Scraping Code:</u> 80 points
<u>Report:</u> 20 points